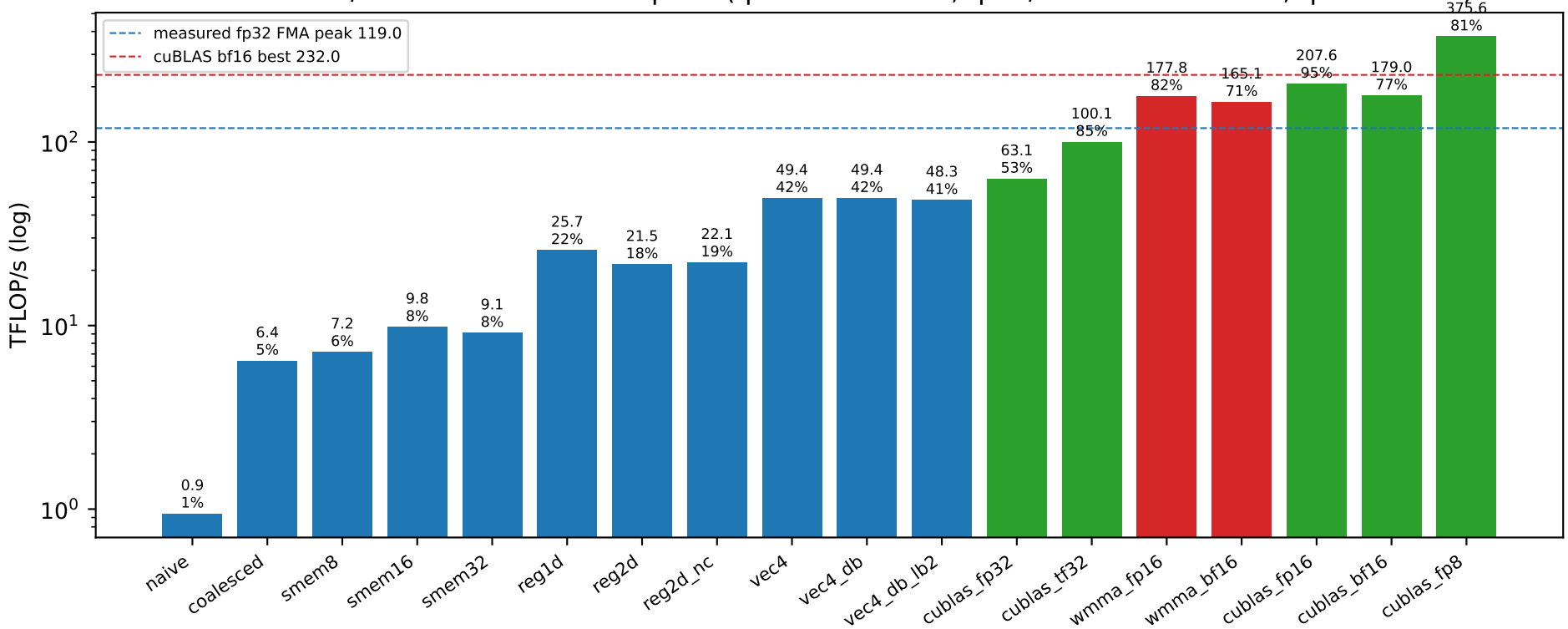
GEMM ladder, M=N=K=4096. blue: my fp32, red: my WMMA, green: cuBLAS
label: TFLOP/s and % of measured peak (fp32: FMA chain; fp16/bf16: best cuBLAS; fp8: 2x bf16)
measured fp32 FMA peak 119.0
cuBLAS bf16 best 232.0
TFLOP/s (log)
10^0
10^1
10^2
0.9
1%
6.4
5%
7.2
6%
9.8
8%
9.1
8%
25.7
22%
21.5
18%
22.1
19%
49.4
42%
49.4
42%
48.3
41%
63.1
53%
100.1
85%
177.8
82%
165.1
71%
207.6
95%
179.0
77%
375.6
81%
naive
coalesced
smem8
smem16
smem32
reg1d
reg2d
reg2d_nc
vec4
vec4_db
vec4_db_lb2
cublas_fp32
cublas_tf32
wmma_fp16
wmma_bf16
cublas_fp16
cublas_bf16
cublas_fp8